The Dirac Equation

Dirac wanted to get an eq$\underline{\underline{n}}$ linear in $\frac{\partial}{\partial t}$. Covariance dictates that such an eq$\underline{\underline{n}}$ must also be linear in $\vec{\nabla}$.

$$\boxed{H\psi = (\vec{\alpha}\cdot\vec{P} + \beta m)\psi}$$

We demand.

$$\boxed{H^2\psi = (\vec{P}^2 + m^2)\psi}$$

$$\therefore H^2\psi = (\alpha_i P_i + \beta m)(\alpha_j P_j + \beta m)\psi$$

$$= (\underbrace{\alpha_i^2}_{1} P_i^2 + \underbrace{(\alpha_i\alpha_j + \alpha_j\alpha_i)}_{0} P_i P_j + \underbrace{(\alpha_i\beta + \beta\alpha_i)}_{0} P_i m + \underbrace{\beta^2}_{1} m^2)\psi$$

$\alpha_i^2 = \beta^2 = 1$ and α_i & β anticommute with each other

$\Rightarrow \alpha_i, \beta, \psi$ can not be numbers, ψ must become a col. matrix.

The lowest-dimensionality $= 4$

Choice of α_i, β is not unique.

The Dirac-Pauli representation:

$$\boxed{\vec{\alpha} = \begin{pmatrix} 0 & \vec{\sigma} \\ \vec{\sigma} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} \mathbb{I}_2 & 0 \\ 0 & -\mathbb{I}_2 \end{pmatrix}}$$

where

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

A four-component column vector ψ which satisfies the Dirac eqn is called a Dirac spinor.

$$i\frac{\partial\psi}{\partial t} = -i\vec{\alpha}\cdot\bar{\nabla}\psi + \beta m\psi.$$

Multiply with β from the left.

$$i\beta\frac{\partial\psi}{\partial t} = -i\beta\vec{\alpha}\cdot\bar{\nabla}\psi + m\psi.$$

This can be written with the Dirac γ matrices

$$i\gamma^0\frac{\partial\psi}{\partial t} + i\gamma^i\frac{\partial\psi}{\partial x^i} - m\psi = 0$$

where $\gamma^0 = \beta$

$\gamma^i = \beta\alpha^i$

or more compactly as $\boxed{(i\gamma^\mu\partial_\mu - m)\psi = 0}$

Openning up: $\displaystyle\sum_{k=1}^{4}\left[\sum_\mu i(\gamma^\mu)_{jk}\partial_\mu - m\delta_{jk}\right]\psi_k = 0$

Test:

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}$$

$(\gamma^0)^2 = \mathbb{I}_4$ (will write just 1 from now on)

$(\gamma^k)^2 = \beta\alpha^k\beta\alpha^k = -1.$

$\gamma^{0\dagger} = \gamma^0$

$\gamma^{k\dagger} = (\beta\alpha^k)^\dagger = \alpha^k\beta = -\beta\alpha^k = -\gamma^k$

or, $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$

What about currents?

For K-G eqn. we needed ψ^* in the current expression.
Here we need hermitian conjugate of ψ.

The hermitian conjugate of the Dirac eqn.

$$\boxed{-i\frac{\partial \psi^+}{\partial t}\gamma^0 - i\frac{\partial \psi^+}{\partial x^k}(-\gamma^k) - m\psi^+ = 0}$$

To restore the covariant form, we multiply the above expression by γ^0 from the right and introduce the adjoint spinor

$\bar{\psi} \equiv \psi^+\gamma^0$ we get

$$-i\frac{\partial \bar{\psi}}{\partial t}\gamma^0 - i\frac{\partial \bar{\psi}}{\partial x^k}\left(-\gamma^0\gamma^k\gamma^0\right) - m\bar{\psi} = 0$$

We can write this compactly as

$$\boxed{i\partial_\mu \bar{\Psi}\gamma^\mu + m\bar{\Psi} = 0} \leftarrow \psi$$

We had

$$\bar{\psi} \longrightarrow \boxed{i\gamma^\mu \partial_\mu \psi - m\psi = 0}$$

$$\bar{\Psi}\gamma^\mu \partial_\mu \psi + (\partial_\mu \bar{\Psi})\gamma^\mu \psi = 0$$

or $\partial_\mu (\bar{\Psi}\gamma^\mu \psi) = 0$

$\therefore$ We can take $\boxed{j^\mu = \bar{\psi}\gamma^\mu \psi}$ Probability current.

$\rho = j^0 = \bar{\Psi}\gamma^0\psi$
$= \psi^\dagger \psi = \sum_{i=1}^{4} |\psi_i|^2$
It is never $-$ve

So actual electron current (e.m.)

$$\boxed{j^\mu = -e\bar{\psi}_e \gamma^\mu \psi_e}$$

So what did we learn?

- We can have an operator linear in $\frac{\partial}{\partial t}$ and covariant. But it will be a matrix eqn.

$$-i\vec{\alpha}\cdot\vec{\nabla}\psi + \beta m \psi = i\frac{\partial \psi}{\partial t}$$

$$\Rightarrow \quad i\gamma^0 \frac{\partial \psi}{\partial t} + i\gamma^i \frac{\partial \psi}{\partial x^i} - m\psi = 0$$

$$\boxed{\left(i\gamma^\mu \partial_\mu - m\right)\psi = 0}$$

(Four coupled equations.)

or $\boxed{\left(\gamma^\mu p_\mu - m\right)\psi = 0}$

Dirac-Pauli rep.

$$\gamma^0 = \beta \rightarrow \begin{pmatrix} \mathbb{I}_2 & 0 \\ 0 & -\mathbb{I}_2 \end{pmatrix}$$

$$\gamma^i = \beta\alpha^i \rightarrow \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

Pauli matrices

$$\psi = \begin{pmatrix} x \\ x \\ x \\ x \end{pmatrix}$$

- We learned about the properties of the γ^μ matrices anti-commutator

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}\mathbb{I}_4$$

where $g^{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$

$$(\gamma^0)^2 = \mathbb{I}_4\ , \quad (\gamma^1)^2 = (\gamma^2)^2 = (\gamma^3)^2 = -\mathbb{I}_4$$

$$(\gamma^0)^\dagger = \gamma^0, \quad (\gamma^i)^\dagger = -\gamma^i \Rightarrow \gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$$

⊙ We learned about the conjugate form of Dirac eqn

$$\boxed{i\partial_\mu\bar{\Psi}\gamma^\mu + m\bar{\Psi} = 0}$$ (it is like Schrödinger eqn written in terms of ψ^*)

where $\bar{\Psi} = \psi^\dagger\gamma^0 = (\times \ \times \ \times \ \times)$

Dirac conjugate

• The conjugate eqn implies

$$\bar{\psi}\gamma^\mu \partial_\mu \psi + (\partial_\mu \bar{\psi})\gamma^\mu \psi = 0$$

or $\partial_\mu(\underbrace{\bar{\psi}\gamma^\mu\psi}_{j^\mu}) = 0$ [Remember, γ^μ is a fixed matrix]

This is similar to the K-G (Schrödinger eqn) case

$$\partial_\mu\Big(\underbrace{i(\phi^* \partial^\mu \phi - \phi\, \partial^\mu \phi^*)}_{j^\mu_{KG}}\Big) = 0$$

j^μ_{KG} ← probability current

Check $j^0 = \bar{\psi}\gamma^0\psi = \psi^\dagger\psi = \sum_{i=1}^{4} |\psi_i|^2 \geq 0$

So we can define the electromagnetic current carried by an electron as

$$\boxed{j^\mu_e = -e\,\bar{\psi}\gamma^\mu\psi}$$ ← this is a 4-vector!

Question: Show the components ψ_i satisfy the K-G equation.

How to solve the Dirac eqn?

Guess: $\psi(x) = u(p) e^{-ip.x}$ $\left\{ u(p) = \begin{pmatrix} \times \\ \times \\ \times \\ \times \end{pmatrix} \text{indep. of } x \right\}$

$\therefore$ We get $(\gamma^\mu p_\mu - m) u(p) = 0$

Feynman introduced the shorthand $\not{a} = \gamma^\mu a_\mu = a^\mu \gamma_\mu$

$\therefore$ $(\not{p} - m) u(p) = 0$

$\Rightarrow$ $(\vec{\alpha} \cdot \vec{p} + \beta m) u = E u$

If the particle is at rest (or if we are moving with the electron)

$\vec{p} = 0 \Rightarrow \begin{pmatrix} m I_2 & 0 \\ 0 & -m I_2 \end{pmatrix} u = E u$

Clearly this has four eigenvectors with two energy eigenvalues

$$\underbrace{\begin{pmatrix}1\\0\\0\\0\end{pmatrix} \rightarrow +m = E \, , \quad \begin{pmatrix}0\\1\\0\\0\end{pmatrix} \rightarrow +m = E}_{E>0 \text{ electrons } (\uparrow\downarrow)} \, , \quad \underbrace{\begin{pmatrix}0\\0\\1\\0\end{pmatrix} \rightarrow -m = E \, , \quad \begin{pmatrix}0\\0\\0\\1\end{pmatrix} \rightarrow -m = E}_{E>0 \text{ positrons } (\downarrow\uparrow)}$$

For $\vec{p} \neq 0$,

$$\begin{pmatrix} m & \vec{\sigma}\cdot\vec{p} \\ \vec{\sigma}\cdot\vec{p} & -m \end{pmatrix} \begin{pmatrix} u_A \\ u_B \end{pmatrix} = E \begin{pmatrix} u_A \\ u_B \end{pmatrix}$$

or $\vec{\sigma}\cdot\vec{p}\, u_B = (E-m)\, u_A$

& $\vec{\sigma}\cdot\vec{p}\, u_A = (E+m)\, u_B$

Let us see if we can retain the ($\vec{p}=0$) solutions for the electron ($E>0$)

i.e., $u_A^{(1)} = \chi^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $u_A^{(2)} = \chi^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$\Rightarrow \quad u_B^{(i)} = \frac{\vec{\sigma}\cdot\vec{p}}{E+m}\,\chi^{(i)}$$

So we get the following two 'spinors' for $E>0$

$$u^{(i)} = N \begin{pmatrix} \chi^{(i)} \\ \dfrac{\vec{\sigma}\cdot\vec{p}}{E+m}\,\chi^{(i)} \end{pmatrix}, \quad E>0 \quad \left[\text{Check } \vec{p}\to 0\right]$$

For the $-$ve E solutions let $u_B^{(i+2)} = \chi^{(i)} \Rightarrow u_A^{(i+2)} = -\dfrac{\vec{\sigma}\cdot\vec{p}}{|E|+m}\,\chi^{(i)}$

$\therefore$ We get

$$u^{(i+2)} = N \begin{pmatrix} \dfrac{-\vec{\sigma}\cdot\vec{p}}{|E|+m}\,\chi^{(i)} \\ \chi^{(i)} \end{pmatrix}, \quad E<0$$

It is easy to verify that the four solutions are orthogonal.

$$u^{(r)+}\, u^{(s)} = 0 \quad \text{if } r \neq s .$$

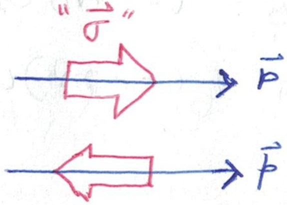
The extra twofold degeneracy implies that we need another observable (operator) that commutes with H & $\vec{p}$ whose eigenvalues can be used for labelling the states.

$$\boxed{\vec{\Sigma}\cdot\hat{p} = \begin{pmatrix} \vec{\sigma}\cdot\hat{p} & 0 \\ 0 & \vec{\sigma}\cdot\hat{p} \end{pmatrix}}$$

Helicity

The "spin" component in the direction of momentum $\dfrac{\vec{p}}{|\vec{p}|} = \hat{p}$ is a "good" quantum number.

$$\lambda = \begin{cases} +\frac{1}{2} & \text{positive helicity} \\ -\frac{1}{2} & \text{negative helicity} \end{cases}$$



{Check: take $\hat{p}$ along $\hat{z}$}

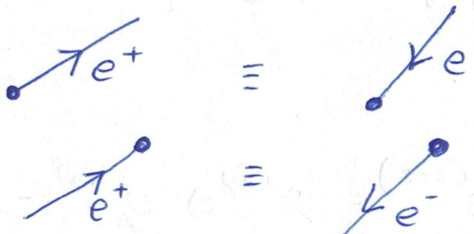
It is convenient to use separate "positron" spinors.

$$u^{(3,4)}(-p)\,e^{-i(-p)\cdot x} \equiv v^{(2,1)}(p)\,e^{ip\cdot x}$$

For an electron of energy $-E$ & momentum $-\vec{p}$ we have

$$(-\not{p} - m)\,u(-p) = 0$$

$$\Rightarrow (\not{p} + m)\,v(p) = 0 \quad (p^0 > 0)$$



spin is also flipped with momentum
$\Rightarrow$ helicity unchanged.

$$p, \Sigma, \lambda \iff -p, -\Sigma, +\lambda$$

$\psi^\dagger \psi$ is not a scalar under Lorentz transformation!

But $\boxed{\bar{\psi}\psi \text{ is.}}$

We also saw $\bar{\psi}\gamma^\mu\psi$ is a 4-vector. Question: How many such "bilinears" are there?

$$(\bar{\psi})(\underbrace{4\times 4}_{?})(\psi)$$

Notation: $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 \quad = \begin{pmatrix} 0 & \mathbb{I}_2 \\ \mathbb{I}_2 & 0 \end{pmatrix}$ (in the Dirac-Pauli rep)

$$\Rightarrow \gamma^{5\dagger} = \gamma^5 \;\&\; (\gamma^5)^2 = \mathbb{I}_4$$

$$\gamma^5\gamma^\mu + \gamma^\mu\gamma^5 = 0 \Rightarrow \gamma^5\gamma^\mu = -\gamma^\mu\gamma^5.$$

Show that if $E \gg m$ (i.e, extreme relativistic limit, $|E| = |p|$)

$$\gamma^5 u^{(1,2)} = \begin{pmatrix} \vec{\sigma}\cdot\hat{p} & 0 \\ 0 & \vec{\sigma}\cdot\hat{p} \end{pmatrix} u^{(1,2)}$$

Bilinears	type	No of components	Space inversion, P
$\bar{\psi}\psi$	scalar	1	(+)
$\bar{\psi}\gamma^\mu\psi$	vector	4	space comp: (−)
$\bar{\psi}\left[\frac{i}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)\right]\psi$	tensor	6	
$\bar{\psi}\gamma^5\gamma^\mu\psi$	axial-vector	4	Space comp: (+)
$\bar{\psi}\gamma^5\psi$	pseudoscalar	1	(−)

γ^5 is called the chirality operator

$$P_R = \frac{1}{2}(1+\gamma^5) \qquad P_i^2 = P_i \; , \; P_i P_j = \delta_{ij} (P_i) \text{ (no sum)}$$

$$P_L = \frac{1}{2}(1-\gamma^5) \qquad P_R + P_L = \mathbb{1}_4 .$$